WESTERN MICHIGAN UNIVERSITY

College of Engineering and Applied Sciences

Course Name: Lab. Intro. To Microprocessors

Course Number: ECE 2510

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Lab #9 prelab

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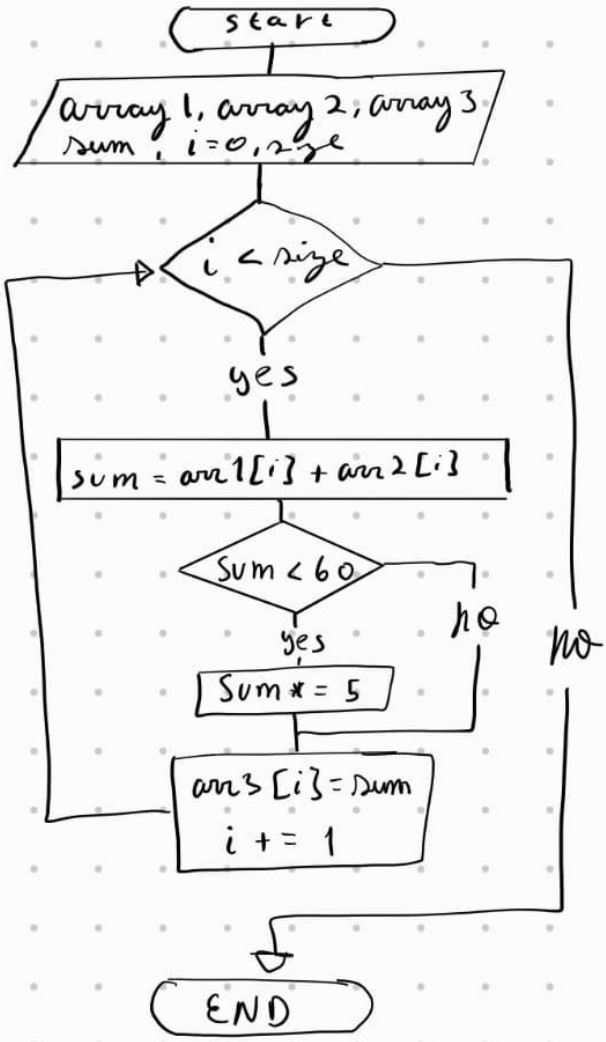
Prelab 9

Task 1

flowchart



pseudo code

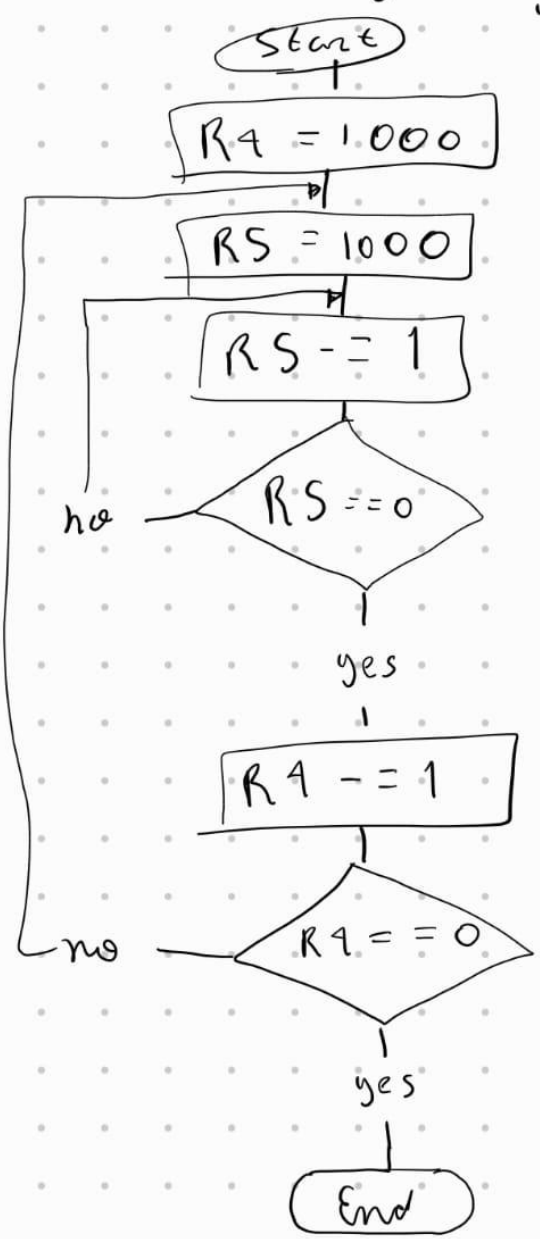
```
algorithm sum do:
Input: arr1, arr2, arr3, size

for i=0 to size do:
    sum = arr1[i] + arr2[i]
    if sum < 60 do:
        sum *= 5
    end if
    arr3[i] = sum
end for
```

Task 2

delay

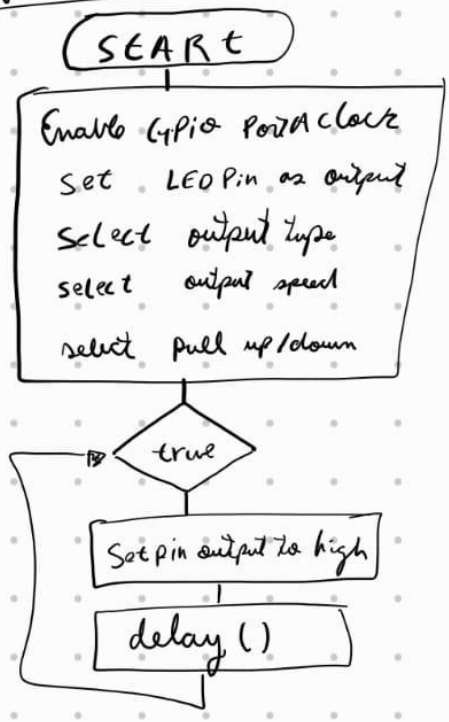
for 1 sec delay
outer = 1000 iter
inner = 1000 iter
at 4MHz



task 2

Blink

flowchart

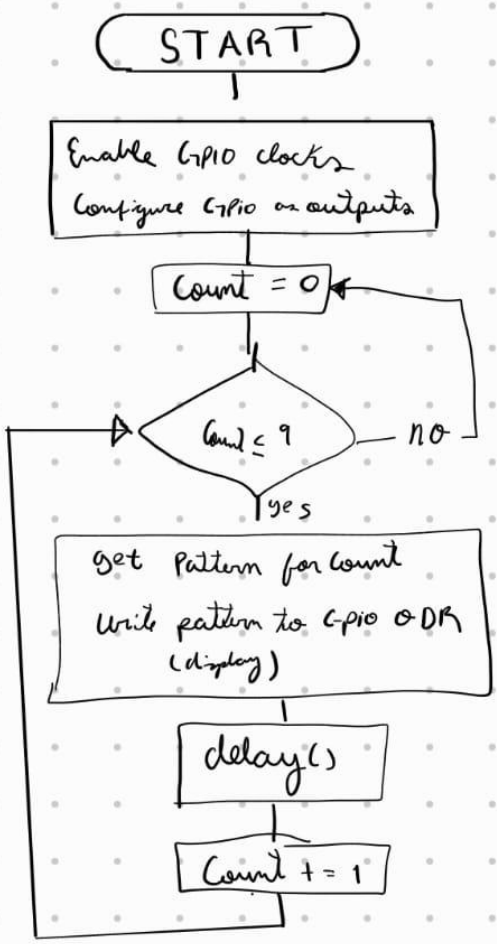


pseudocode

```
algorithm blink do
  setup LED Pin ();
  while true do:
      turn on LED ;
      delay ();
  end loop
```

Task 3

flowchart



Pseudocode

```
algorithm Counter:
    Enable clock for GPIO Port
    Configure segment Pins
    Configure digit select pin

    define array of patterns
     [Hex0, Hex1, . . . , Hex9]
    set count = 0

    while true do:

        Current_Number = patterns[Count]

        select digit
        Write current shape and selected
         digit to ODR

        delay ()

        Count += 1

        if count > 9 then:
            count = 0
        end if

    end loop
```